

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture

- Proposition p : An statement which is **either** T or F
- New propositions given p and q :
 - Negation $\neg p$
 - Conjunction (AND) $p \wedge q$
 - Disjunction (OR) $p \vee q$
 - Exclusive disjunction (XOR) $p \oplus q$
 - Implication $p \rightarrow q$
 - Biconditional $p \leftrightarrow q$
- $p \rightarrow q$ is read as:
 - if p , then q
 - p is sufficient for q
- $p \leftrightarrow q$ is read as:
 - if p , then q and conversely
 - p if and only if q

Some definitions:

$\leftrightarrow$ iff

Given the implication statement $p \rightarrow q$, we can build the following new statements:

- **Converse:** $q \rightarrow p$
- **Contrapositive:** $\neg q \rightarrow \neg p$
- **Inverse:** $\neg p \rightarrow \neg q$

Question:

Show that $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are equivalent, that is, both have the same truth values: $p \rightarrow q \equiv \neg q \rightarrow \neg p$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$\neg p$	$\neg q$	$\neg q \rightarrow \neg p$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

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T	T	T
T	F	F
F	T	T
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p	q	$\neg p$	$\neg q$	$\neg q \rightarrow \neg p$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Logical equivalence

Example. Show that $\underbrace{p \vee (q \wedge r)}_{\text{LHS}} \equiv \underbrace{(p \vee q) \wedge (p \vee r)}_{\text{RHS}}.$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$


Logical equivalence

Example. Show that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Logical equivalence

Example. Show that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

The number of rows grows exponentially with the number of propositions involved.
Very cumbersome.

Logical equivalence

A systemic approach for showing equivalence.

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Establish some laws:

- Identity laws: $p \wedge T \equiv p$ and $p \vee F \equiv p$
 - Domination laws: $p \wedge F \equiv F$ and $p \vee T \equiv T \rightarrow \text{lazy eval.}$
 - Idempotent laws: $p \wedge p \equiv p$ and $p \vee p \equiv p$
 - Double negation: $\neg(\neg p) \equiv p$
 - Commutative laws: $p \vee q \equiv q \vee p$ and $p \wedge q \equiv q \wedge p$
 - Negation laws: $p \vee \neg p \equiv T$ and $p \wedge \neg p \equiv F$
 - Associative laws: $(p \vee q) \vee r \equiv p \vee (q \vee r)$ and $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- 

Logical equivalence

Some more laws:

- Distributive laws:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \quad \text{and} \quad p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

- De Morgan's laws:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q \quad \text{and} \quad \neg(p \wedge q) \equiv \neg p \vee \neg q$$

- Absorption laws:

$$p \vee (p \wedge q) \equiv p \quad \text{and} \quad p \wedge (p \vee q) \equiv p$$

- Implication law:

$$p \rightarrow q \equiv \neg p \vee q$$

Logical equivalence

$$\boxed{p \rightarrow q \equiv \neg p \vee q}$$

Example. Show that $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

$$p \rightarrow q \equiv \neg p \vee q \quad \text{imp. l.}$$

$$\equiv (\neg q) \vee (\neg p) \quad \text{e. l.}$$

$$\equiv \neg (\neg q) \vee \neg p$$

$$\equiv \neg q \rightarrow \neg p$$

double neg.

imp. l.

QED

Logical equivalence

$$\boxed{p \rightarrow q \equiv \neg p \vee q}$$

Example. Show that $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$.

$$\begin{aligned} (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) * \\ &\equiv (r \vee \neg p) \wedge (r \vee \neg q) \\ &\equiv r \vee (\neg p \wedge \neg q) \uparrow \\ &\equiv r \vee \neg(p \vee q) \uparrow \\ &\equiv \underbrace{\neg(p \vee q)} \vee \underbrace{r} \\ &\equiv (p \vee q) \rightarrow r \end{aligned}$$

□

Logical equivalence

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

Example. Show that $p \leftrightarrow q \equiv (\neg p \wedge \neg q) \vee (p \wedge q)$.

$$\begin{aligned} p \leftrightarrow q &\equiv (p \rightarrow q) \wedge (q \rightarrow p) \\ &\equiv (\neg p \vee q) \wedge (\neg q \vee p) \\ &\equiv ((\neg p \vee q) \wedge \neg q) \vee ((\neg p \vee q) \wedge p) \\ \text{absorb. laws} &\equiv (\neg p \wedge \neg q) \vee (q \wedge p) \end{aligned}$$

Tautology

Definition

A proposition that is always true, regardless of the truth value of the involved propositions, is called **tautology**.

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Example. $p \vee \neg p$ is a tautology.

Question

When is $p \leftrightarrow q$ a tautology?

Tautology

Example. Determine whether $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is a tautology.

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It is a tautology. Note that we need to show that $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is always true, regardless of the truth values of p and q . To do so, we use equivalence laws as follows:

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p \equiv \neg p \vee \neg(\neg q \wedge (p \rightarrow q)) \quad \text{(implication)}$$

$$\equiv \neg p \vee (q \vee \neg(p \rightarrow q)) \quad \text{(De Morgan)}$$

$$\equiv \neg p \vee (q \vee (p \wedge \neg q)) \quad \text{(De Morgan, implication)}$$

$$\equiv (\neg p \vee q) \vee (p \wedge \neg q) \quad \text{(associative)}$$

$$\equiv \neg(p \wedge \neg q) \vee (p \wedge \neg q) \quad \text{(De Morgan)}$$

$$\equiv T \quad \text{(negation)}$$

Tautology

Example. Determine whether $(\neg p \wedge (p \rightarrow q)) \rightarrow p$ is a tautology.

Tautology

Example. Determine whether $(\neg p \wedge (p \rightarrow q)) \rightarrow p$ is a tautology.

No. We need to pick a particular truth values for p and q making the proposition false. To see this, recall that the implication is false when its hypothesis is true and its conclusion is false. So, we want p to be false.

Now, let's look at the hypothesis, i.e., $(\neg p \wedge (p \rightarrow q))$. Since p is false, both $\neg p$ and $p \rightarrow q$ are true, thus the hypothesis is true.

$$(\neg F \wedge (F \rightarrow q)) \rightarrow F$$

Thus, if p is false, then the hypothesis is true while the conclusion is false, implying that the implication $(\neg p \wedge (p \rightarrow q)) \rightarrow p$ is false. So it can't be a tautology.

① try logical equivalence
if not tautology

② when is the statement False